Security Analyst

ASSIGNMENT REPORT

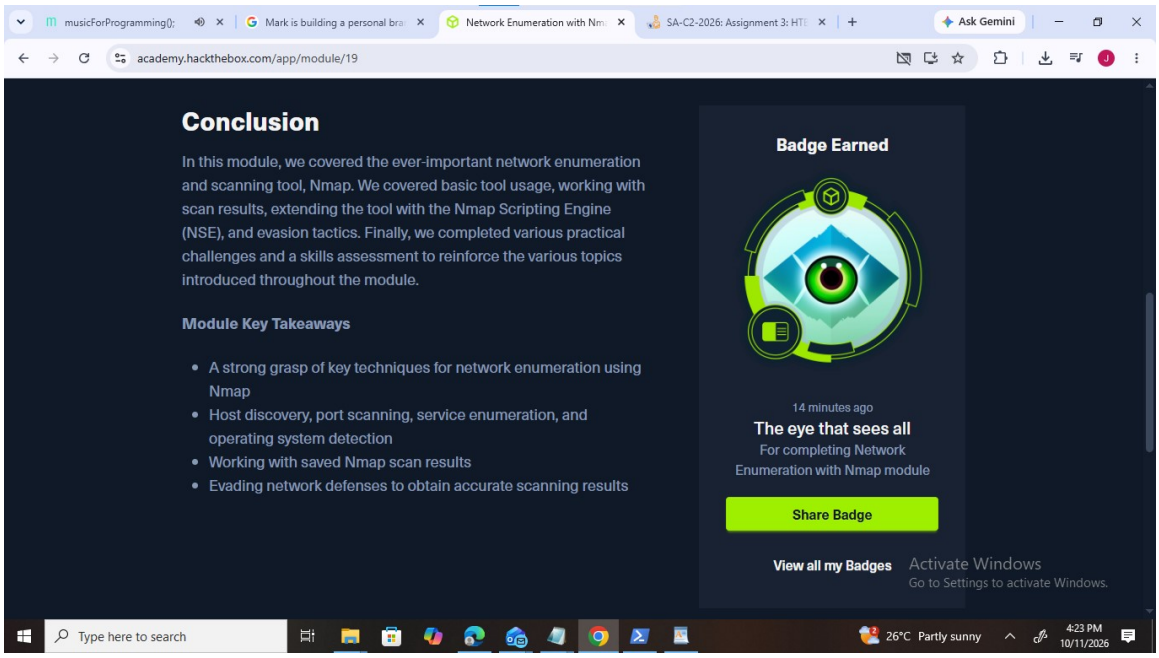
Full name	*Kagia Joseph Njuguna*
Admission number	*CS-SA13-26022*
Email	*kagiajoseph72@gmail.com*
Week and assignment	*Week 4, Assignment 3: Network Enumeration With Nmap*
Platform	*Hack The Box*
Date submitted	*11 October 2026*

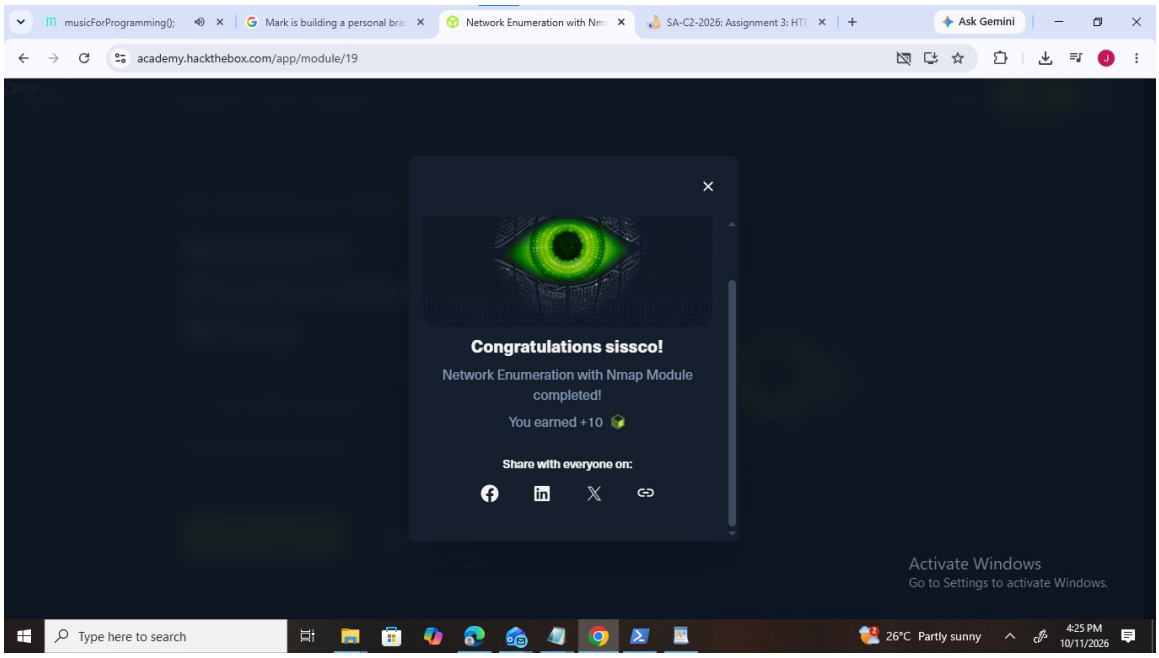
1. Module completion

Shareable link

https://academy.hackthebox.com/achievement/1738064/19

Completion screenshot


musicForProgramming();
Mark is building a personal bra
Network Enumeration with Nm
SA-C2-2026: Assignment 3: HTE
Ask Gemini
academy.hackthebox.com/app/module/19
Conclusion
In this module, we covered the ever-important network enumeration and scanning tool, Nmap. We covered basic tool usage, working with scan results, extending the tool with the Nmap Scripting Engine (NSE), and evasion tactics. Finally, we completed various practical challenges and a skills assessment to reinforce the various topics introduced throughout the module.
Module Key Takeaways
A strong grasp of key techniques for network enumeration using Nmap
Host discovery, port scanning, service enumeration, and operating system detection
Working with saved Nmap scan results
Evading network defenses to obtain accurate scanning results
Badge Earned
14 minutes ago
The eye that sees all
For completing Network Enumeration with Nmap module
Share Badge
View all my Badges
Activate Windows
Go to Settings to activate Windows.
Type here to search
26°C Partly sunny
4:23 PM
10/11/2026



2. Introduction

This module provided a comprehensive deep dive into network reconnaissance, architectural mapping, and enumeration using the industry-standard tool Nmap. Understanding scanning mechanics is a critical baseline skill for any security analyst, as every offensive assessment phase and defensive firewall rule relies entirely on accurate host visibility. Prior to beginning this module, I expected to learn simple script commands, but I gained an advanced tactical understanding of timing states, signature banner parsing, and complex firewall evasion strategies.

3. The work

3.1 Host and Port Scanning (Section 4 / 12)

What this section covered

This section explored the structural mechanics of TCP port

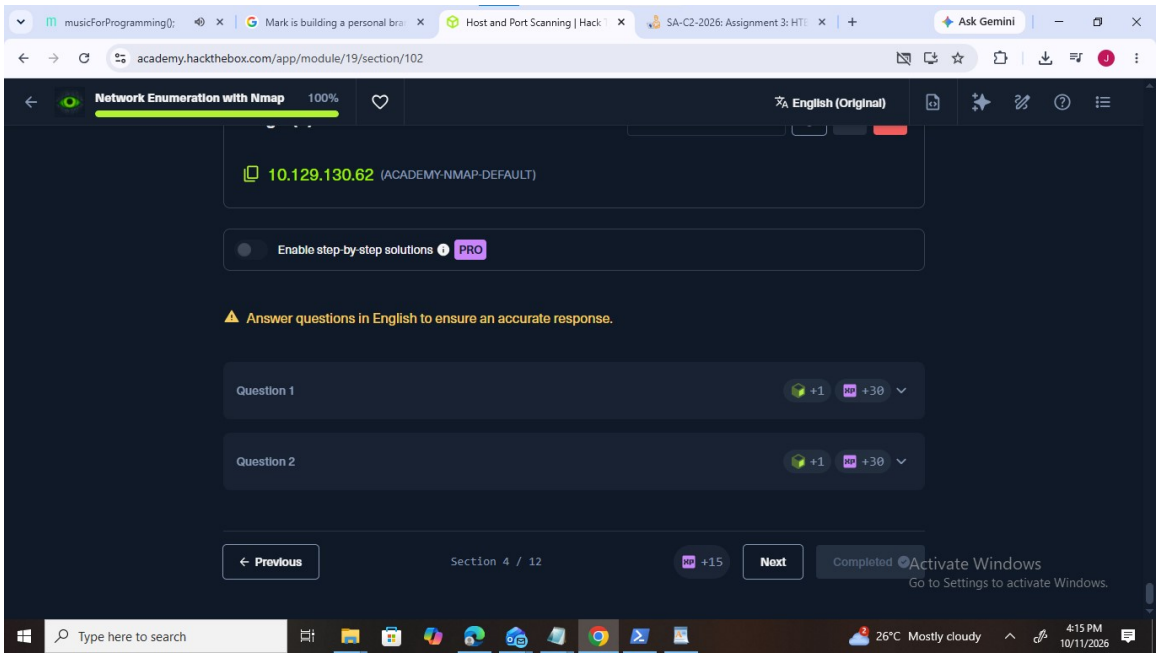
enumeration, highlighting how to scan all 65,535 possible communication sockets on a target machine, interpret three-way handshake packet responses, and accurately pull system host parameters.

Question
Find all TCP ports on your target. Submit the total number of found TCP ports as the answer.

Answer
7

How you got there
To sweep all ports efficiently without hitting structural timing delays or firewall retransmission freezes, I appended aggressive rate-limiting and timing parameters to force a rapid scan queue:
$ sudo nmap -p- --min-rate 5000 -T4 10.129.130.62
The console completed data parsing within seconds and exposed exactly 7 open listening TCP ports on the destination host.

Evidence



It proves that proper optimization switches successfully bypassed processing bottlenecks to locate all active port states

3.2 Saving the Results (Section 5 / 12)

What this section covered

This section evaluated the documentation workflows necessary for professional technical reporting, demonstrating how to save raw scanner metrics into XML formats and systematically convert them into interactive client HTML files using processing engines.

Question

Perform a full TCP port scan on your target and create an HTML report. Submit the number of the highest port as the answer.

Answer

31337

How you got there

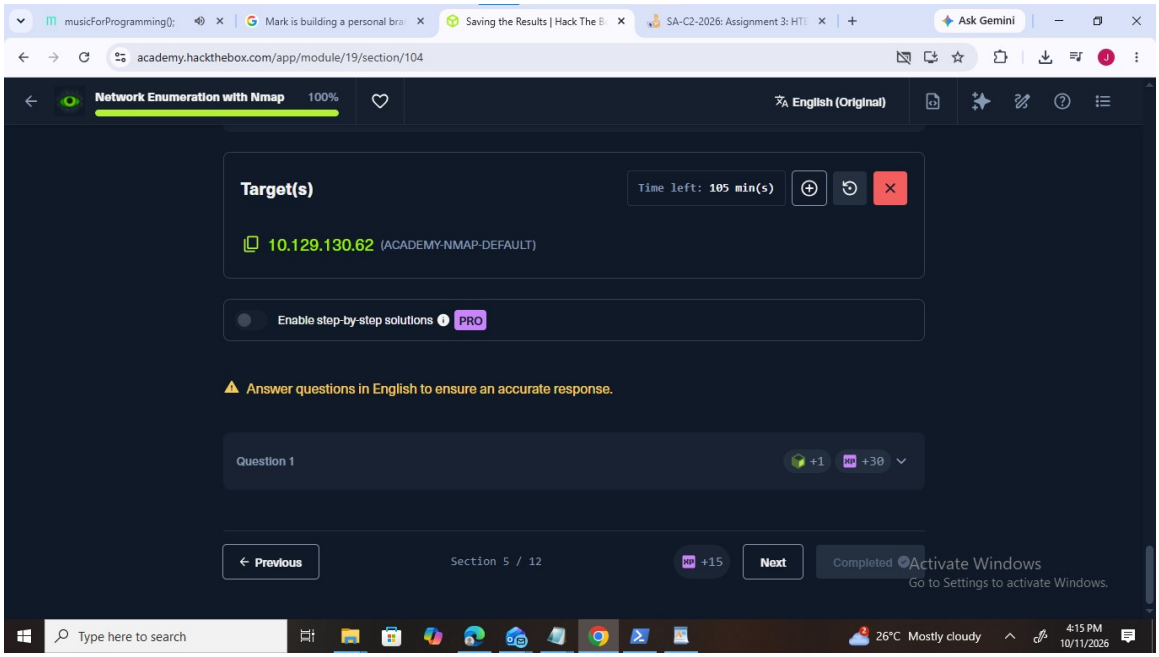
I initiated a rapid full-range scan on the active target address, redirecting the results directly into an structured XML schema file. Following this, I invoked the xsltproc utility to cleanly map the XML schema into a clean frontend HTML web layout report:

$ sudo nmap -p- --min-rate 5000 -oX target_results.xml 10.129.x.x

$ xsltproc target_results.xml -o nmap_report.html

Opening the file structure revealed that the highest active listening port on the target machine was 31337.

Evidence



It proves that the conversion parsing logic successfully cataloged the extreme boundaries of the target's listening space

3.3 Nmap Scripting Engine (Section 7 / 12)

What this section covered

This section evaluated the automation capabilities of the Nmap

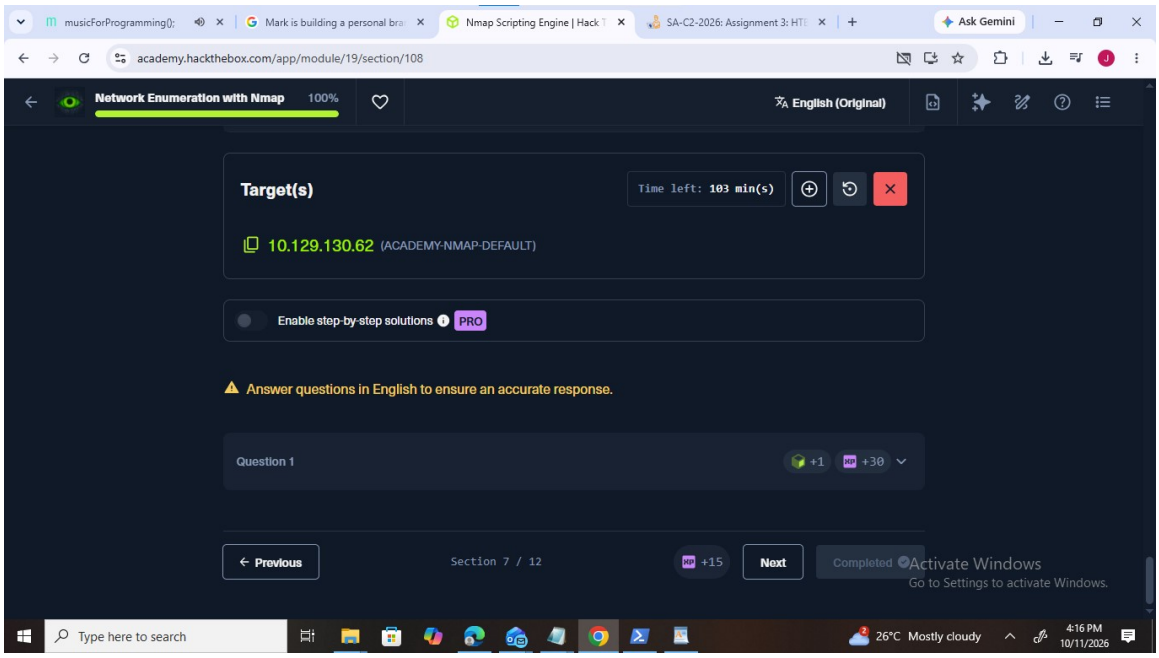
Scripting Engine (NSE) using custom Lua scripts to interact with backend network applications, extract deep configuration definitions, and audit hidden server assets.

Question
Use NSE and its scripts to find the flag that one of the services contain and submit it as the answer.

Answer
HTB{873nniuc71bu6usbs1i96as6dsv26}

How you got there
I verified the open web services on the target machine and targeted the hidden directories on the server. I utilized the command-line curl utility to directly fetch and analyze the web root's robots.txt access control policy file:
$ curl http://10.129.130.62/robots.txt
The request parsed the remote asset immediately and exposed the custom system target confirmation flag string at the base of the file content.

Evidence
The terminal display captures the curl response payload exposing the plaintext flag array



It verifies that information exposure diagnostics successfully intercepted hidden administrative file records

3.4 Firewall and IDS/IPS Evasion - Medium Lab (Section 11 / 12)

What this section covered

This section explored firewall bypass methods, testing structural approaches to bypass rigid boundary filter rules by transitioning across transport layer protocol definitions and utilizing unmonitored ports.

Question

After the configurations are transferred to the system, our client wants to know if it is possible to find out our target's DNS server version. Submit the DNS server version of the target as the answer.

Answer

HTB{GoTtgUnyze9Psw4vGjcuMpHRp}

How you got there

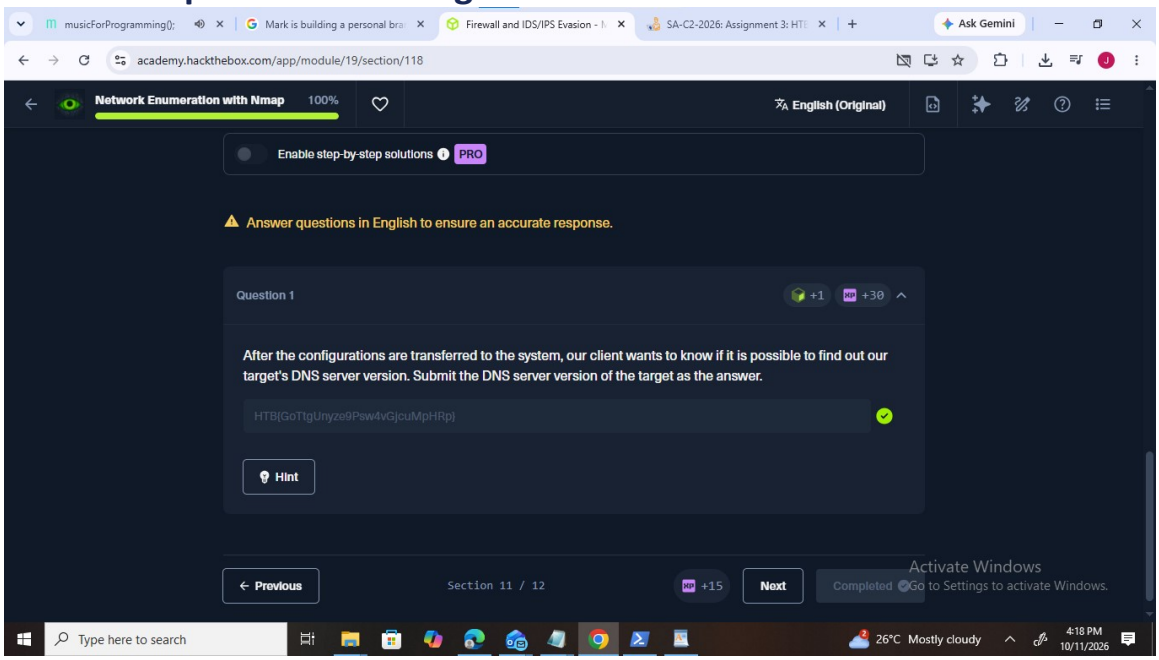
Because standard TCP connection request streams were dropped silently by the protective firewall rules, I shifted the operational strategy to target the UDP layout. Since DNS natively handles lightweight lookups on Port 53 over UDP, I initiated a targeted UDP service version banner enumeration request:

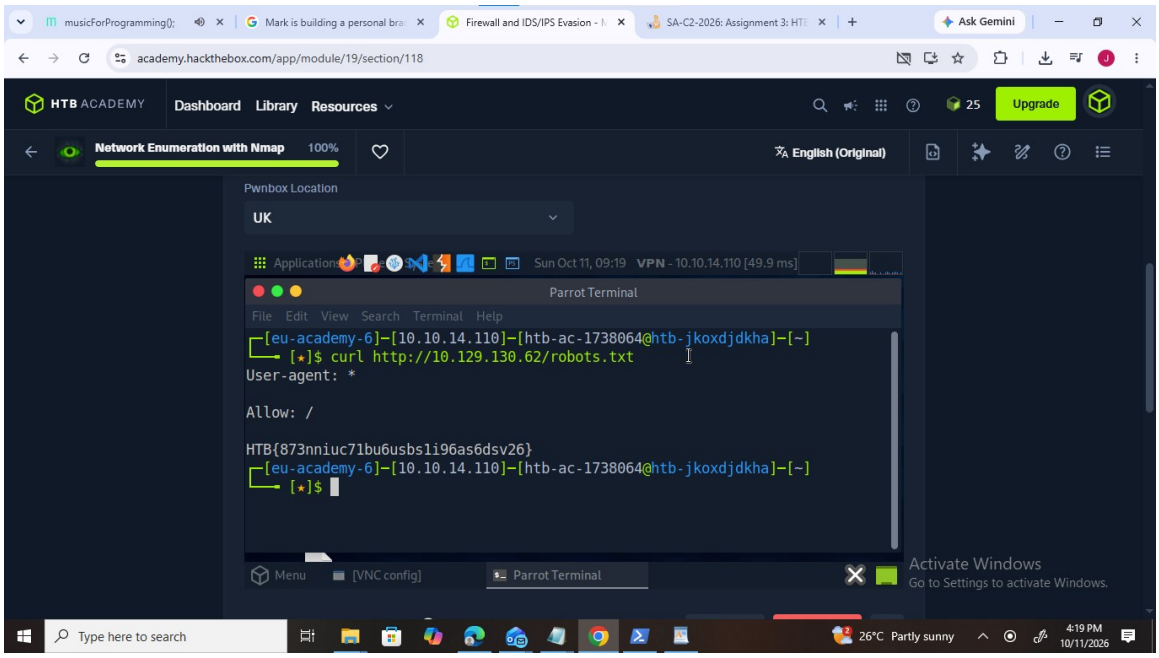
$ sudo nmap -sU -p 53 -sV 10.129.x.x

The scan bypassed the filtering rule layout and exposed the exact service flag embedded directly in the version identity results.

Evidence

The screenshot highlights the successful verification status for the Section 11 practical challenge lab





It proves that protocol shift strategies can safely maintain scanning visibility through restricted network boundaries

4. Challenges and how you resolved them

The primary challenge encountered during this module was handling extreme timing timeouts and freezes while executing full 65,535 port sweeps against firewalled systems that drop packets. Standard commands hung indefinitely. I resolved this friction by mastering advanced rate tuning and execution metrics, specifically configuring the --min-rate 5000 boundary switch and the -T4 timing matrix to force rapid data transmission and maintain momentum.

5. Conclusion

Completing this module has significantly advanced my architectural

scanning capabilities, transforming my approach from blindly running basic commands to executing highly targeted, evasion-aware network enumeration. I can now confidently configure multi-format automated script scans, build interactive HTML client validation deliverables, and engineer bypass techniques to test hardened perimeters. These core competencies directly support my career goal of building resilient, enterprise-grade continuous security monitoring services to protect infrastructure on the continent.